GCMC Simulation

Tuesday, 1 February 2022 11:09 AM

The grand canonical ensemble represents an open system.
The system exchanges molecules with a reservoir at constant chemical potential

$$\text{Partition function} = \Xi(\mu, V, T) = \sum_{N=0}^{\infty} \frac{e^{\beta\mu N}}{N!\,\Lambda^{3N}} \int\cdots\int e^{-\beta U(\vec{r}^N)}\, d\vec{r}^N$$

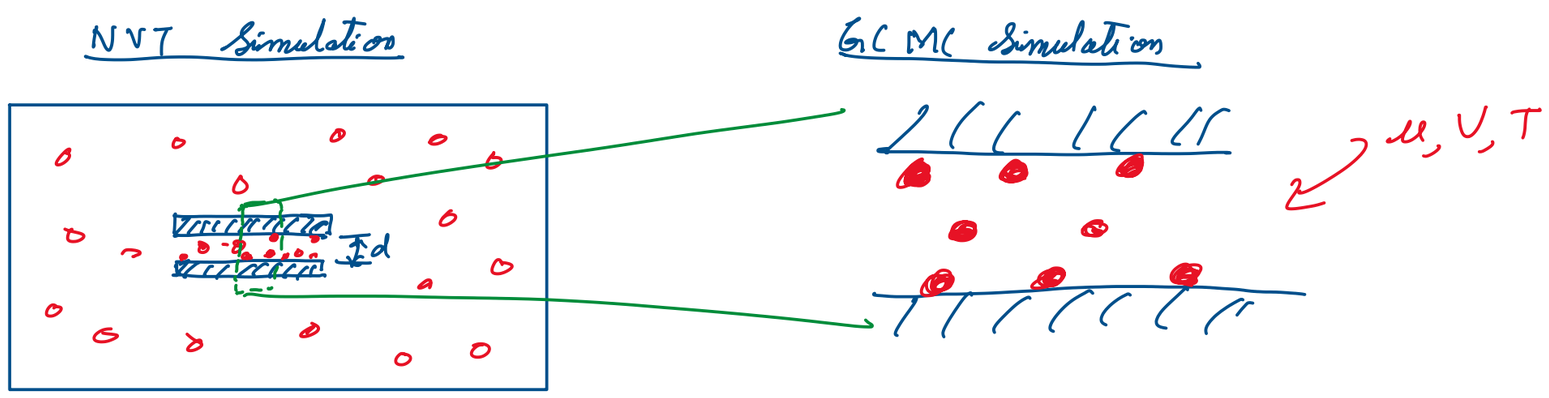
$$= \sum_{N=0}^{\infty} \frac{V^N e^{\beta\mu N}}{N!\,\Lambda^{3N}} \int\cdots\int e^{-\beta U}\, d\vec{s}^N \qquad \vec{s} = \frac{\vec{r}}{V^{1/3}}$$

$$\text{Probability of microstate } f(\vec{r}_1, .., \vec{r}_N; N) \propto \frac{V^N e^{\beta\mu N}}{N!\,\Lambda^{3N}} e^{-\beta U(\vec{r}^N)}$$

Application of GCMC

Adsorption of gas in a pore

Consider a slit pore.



Difficulties with NVT:
1) The entire adsorbent (usually macroscopic in size) has to be simulated
2) One has to simulate the external gas

The GCMC simulation is computationally much cheaper.

Types of monte carlo moves

1) Molecule displacement (similar to move in NVT simulation)

As derived earlier
$$acc(o \to n) = \min\left\{1, e^{-\beta\Delta U}\right\}$$

2) Molecule insertion $(N \to N+1)$

detailed balance
$$f(o)\, \propto(o \to n)\, acc(o \to n) = f(n)\, \propto(n \to o)\, acc(n \to o)$$

$$f(o) \propto \frac{V^N e^{\beta\mu N}}{N!\,\Lambda^{3N}} e^{-\beta U(o)}$$

$$f(n) \propto \frac{V^{N+1} e^{\beta\mu(N+1)}}{(N+1)!\,\Lambda^{3(N+1)}} e^{-\beta U(n)}$$

if the probability of attempting the insertion move $(N \to N+1)$ is equal to the probability of attempting the deletion move $(N+1 \to N)$ then
$$\propto(o \to n) = \propto(n \to o)$$

$$\therefore \frac{acc(o \to n)}{acc(n \to o)} = \frac{f(n)}{f(o)} = \frac{V e^{\beta\mu}}{(N+1)\Lambda^3} e^{-\beta\Delta U} \qquad \Delta U = U(n) - U(o)$$

$$\therefore acc(o \to n) = \min\left\{1, \frac{V e^{\beta\mu N}}{(N+1)\Lambda^3} e^{-\beta\Delta U}\right\}$$

Recall $\beta\mu = \ln \rho\Lambda^3 + \beta\mu^{ex}$

$$\therefore \frac{e^{\beta\mu}}{\Lambda^3} = \rho e^{\beta\mu^{ex}} = z \equiv \text{activity}$$

$$\therefore acc(o \to n) = \min\left\{1, \frac{zV}{N+1} e^{-\beta\Delta U}\right\}$$

Recall definition of fugacity

$$k_B T \ln\left(\frac{f}{P}\right) = \mu(T,P) - \mu^{(id)}(T,P)$$

$$\therefore \ln\left(\frac{f}{P}\right) = \beta\mu - \ln\left(\frac{P\Lambda^3}{k_B T}\right)$$

$$\therefore \frac{f}{P} = e^{\beta\mu}\,\frac{k_B T}{P\Lambda^3}$$

$$\therefore \frac{e^{\beta\mu}}{\Lambda^3} = \beta f$$

$$\therefore \text{acc}(o \to n) = \min\left\{1,\ \frac{\beta f V}{N+1}\, e^{-\beta\Delta U}\right\}$$

3) Molecule deletion $(N \to N-1)$

$$f(o) \propto \frac{V^N e^{\beta\mu N}}{N!\,\Lambda^{3N}}\, e^{-\beta U(o)}$$

$$f(n) \propto \frac{V^{N-1}}{(N-1)!}\,\frac{e^{\beta\mu(N-1)}}{\Lambda^{3(N-1)}}\, e^{-\beta U(n)}$$

after applying detailed balance condition we get

$$\frac{\text{acc}(o \to n)}{\text{acc}(n \to o)} = \frac{N\Lambda^3}{V e^{\beta\mu}}\, e^{-\beta\Delta U}$$

$$= \frac{N}{\beta V}\, e^{-\beta\Delta U}$$

$$= \frac{N}{\beta f V}\, e^{-\beta\Delta U}$$

$$\text{acc}(o \to n) = \min\left\{1,\ \frac{N}{\beta V}\, e^{-\beta\Delta U}\right\}$$

$$= \min\left\{1,\ \frac{N}{\beta f V}\, e^{-\beta\Delta U}\right\}$$

Algorithm for insertion move

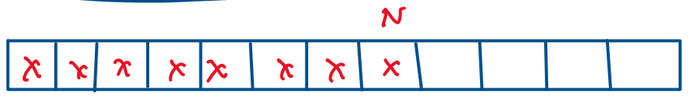
1) In this move, a random point inside the simulation box is chosen uniformly

2) Calculate ΔU

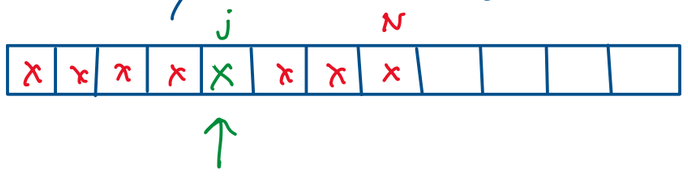
3) if move is accepted update N
   i.e. $N = N+1$

Algorithm for deletion move

Coordinate array



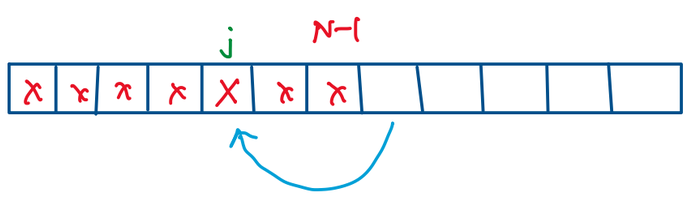
1) Choose a molecule/atom to be deleted at random



2) Compute ΔU

3) if move is accepted then





$$rx(j) = rx(N)$$
$$ry(j) = ry(N)$$
$$rz(j) = rz(N)$$

$$N = N-1$$

GCMC algorithm

1) Specify $\mu(or\ f, z)$, V, T

2) Read initial coordinates
   or
   generate initial coordinates

3) Specify length of simulation, Δr_m, random number seed

4) Perform (nequil + ncycles) number of moves

```
do l = 1, (nequil + ncycles)
    cycle length = max (10, N)
    do m = 1, cycle length
        Choose the move type
        xrand = Ran2 (iseed)
        if (xrand < 0.33) then
            Perform translation move
        else if (xrand < 0.67) then
            Perform insertion move
        else
            Perform the deletion move
        endif

        Sample properties of interest
    enddo
    Optimize value of Δrm during equilibration phase
enddo
```

} each move is chosen with a probability of 1/3